Assignment 04

Question 1
Consider the relation

EMP_PROJ_DEPT(EmpID, EmpName, EmpAddress, DeptID, DeptName, ManagerID, ProjID, ProjName, ProjLocation, Hours)

Assume the following real-world constraints:

- Each employee belongs to exactly one department.
- Each department has exactly one manager.
- Each project has one project name and one project location.
- An employee can work on many projects.
- A project can have many employees.
- Hours depends on the employee-project combination.

Required –

1. Identify all meaningful functional dependencies.
2. Find the candidate key(s).
3. Identify all partial dependencies and transitive dependencies.
4. Explain insertion, deletion, and modification anomalies.
5. Decompose the relation into well-designed relations and justify the design using the informal design guidelines.

Question 2
A university stores examination data in one relation.

EXAM_RECORD(StudentID, StudentName, DegreeProgram, CourseCode, CourseTitle, LecturerID, LecturerName, ExamHall, HallCapacity, Marks, Grade)

Assume,

StudentID -> StudentName, DegreeProgram

CourseCode -> CourseTitle, LecturerID

LecturerID -> LecturerName

ExamHall -> HallCapacity

(StudentID, CourseCode) -> Marks, Grade, ExamHall

Required –

1. Analyze the relation according to all four informal relational design guidelines: semantics of attributes, redundant information and update anomalies, null value problems, and spurious tuple risk.
2. Identify the candidate key.
3. Identify all partial dependencies and transitive dependencies.
4. Decompose the relation into suitable relations.
5. Explain how your decomposition improves the original design.